Student Dropout Prediction

Feature Selection & Feature Engineering Summary

This document explains, in plain terms, which columns from the original dataset were used to train the model, which ones were removed, and which new features were created and why.

Dataset Overview

The dataset contains 200,000 records and 22 original columns. It captures student demographics, course registration details, and assessment activity, with the goal of predicting the student's final result: Pass, Fail, Withdrawn, or Distinction.

Columns Used to Train the Model

Column	What it represents
score, weight	Marks obtained in an assessment, and how much that assessment counts toward the final result
code_module, code_presentation, assessment_type	Which course, which semester/term, and what type of assessment
gender, region, highest_education, imd_band, age_band, disability	Student background and demographic information
num_of_prev_attempts, studied_credits	How many times the student has attempted before, and how many credits they are studying
total_clicks	How actively the student is engaging with the online course platform
date_registration, date_unregistration	When the student registered or unregistered, relative to the course start date

Columns Removed

- id_assessment, id_student — These are just identifier numbers and carry no predictive meaning, so they were dropped.
- Raw date columns (date, date_submitted, date_registration, date_unregistration) — Instead of using these directly, they were used to calculate more meaningful features (explained below), and the original raw columns were then removed.
- Original text-based category columns — Once these were converted into a numeric format the model can use (one-hot encoding), the original text versions were dropped to avoid duplication.

New Features Added (Feature Engineering)

The following features do not exist in the raw data. They were calculated from existing columns to better capture student behavior:

Feature	How it's calculated	What it captures
time_to_submit	date_submitted - date	Whether the assessment was submitted early or late
assessment_late	1 if time_to_submit > 0, else 0	A simple flag for late submission
assessment_early	1 if time_to_submit < 0, else 0	A simple flag for early submission
registration_days_before_start	Absolute value of date_registration (if negative)	How many days before the course started the student registered
clicks_per_day_active	total_clicks / (date + 1)	Average daily engagement with the course platform

Why These Features Were Added

Raw dates and click counts, on their own, don't reveal much of a pattern. For example, knowing a student had 1,000 total clicks says very little by itself — but converting that into clicks per day active clearly shows whether the student was consistently engaged or not. Similarly, instead of leaving the model to work out submission timing from raw dates, a simple late/early flag makes that pattern much easier for the model to learn from.

Model Result

Using this set of features, a Random Forest Classifier was trained on an 80/20 train-test split. The model achieved an overall accuracy of 84% on the test set.

Note: This document is a plain-language summary intended for quick reference. For the full technical workflow, refer to the project notebook (ML_project_2.ipynb).